

# Sprint 49: Judge and Scoring Research - Shruti Kumari

## MT Judge Evaluation: GLM vs Qwen | Translation Quality Evaluation

### Executive Summary

This project tested whether large language models can serve as automated judges of English→Chinese machine translation quality, using FLORES-200 sentence pairs for two language directions: **English→Mandarin** and **English→Cantonese**. Two candidate judge models were evaluated **GLM-4 (9B)** and **Qwen2.5 (7B)**, both run locally and free of cost via Ollama on the full 200-sentence set for each language.

Each judge scored every sentence two ways: a **flat 0–100 holistic score** (the conventional single-number approach) and a **severity-weighted score**, computed deterministically from a structured list of errors the judge tagged as minor, major, or critical. A **reference-free automatic metric** (LaBSE cross-lingual semantic similarity, used as a free stand-in for COMET-QE/CometKiwi) was run alongside both judges as an independent, non-LLM signal.

### **Headline findings:**

- **Flat scoring is nearly uninformative on this data.** Both judges compress almost every sentence into a narrow 85–100 band (std. dev.  $\approx 1.2$ –3.0 points), regardless of how many real errors were found. Severity-weighted scoring preserves that same ceiling for clean sentences but drops meaningfully (down to 65–90) when major or critical errors are present, and — unlike the flat score — comes with an auditable, human-readable reason for every point lost.
- **A single judge misses real errors that a second judge catches — this effect is larger for Cantonese than Mandarin.** Combining GLM and Qwen roughly **doubles** the share of sentences flagged with a major/critical error versus using either judge alone (Mandarin: 5.5%→12%; Cantonese: 10.5%→20%). The only two **critical-severity** errors found anywhere in the whole study were both in Cantonese, and both were caught by Qwen only — GLM never once used the “critical” label across 400 sentences.
- **GLM and Qwen agree with each other only moderately**, and agree *less* on Cantonese than on Mandarin (severity-score correlation: Mandarin  $r=0.35$  vs. Cantonese  $r=0.23$ ), consistent with Cantonese being a lower-resource language for both models.
- **The automatic metric’s coverage was unreliable** — it only successfully scored 8–40% of sentences per file due to an unresolved runtime issue (see Limitations), so its agreement/disagreement numbers should be read as directional, not conclusive.
- **No human-labeled reference data was available** for this round, so judge-vs-human comparison (requested “where possible” in the acceptance criteria) could not be performed. This is the top-priority gap for the next iteration.

**Bottom line recommendation:** use **severity-weighted scoring** as the primary score going forward for both languages; use **GLM alone** as the default judge for Mandarin (cheap, well-calibrated, acceptable miss rate); use **GLM + Qwen together (multi-judge)** as the default for Cantonese, where a single judge — either one — misses meaningfully more real errors. Full reasoning in the Final Recommendation section.

## **Requirements and Acceptance Criteria**

The project’s original acceptance criteria were:

1. Test GLM as a judge on English→Mandarin translations and compare its scores against the current judge.
2. Test GLM as a judge on English→Cantonese translations and compare its scores against the current judge and, where possible, human review.
3. Test a combined multi-judge setup (two or three judges scored together) against the current single-judge setup, on the same set of translations.
4. Record whether the multi-judge setup catches errors — especially Cantonese-specific ones — that a single judge misses.
5. Design and test a severity-weighted scoring approach, distinguishing critical errors from minor ones, compared against the current flat score.
6. Add at least one reference-free automatic metric (e.g. COMET or chrF++) alongside the LLM judge score, and note where it agrees or disagrees with the judge.
7. Document one clear final recommendation: which judge(s) to use, for which language direction, and which scoring method to use going forward.

## **Experimental Setup**

**Data.** FLORES-200 English→Chinese sentence pairs, 200 rows each for Mandarin (Simplified) and Cantonese (Hong Kong written vernacular). Each row is one English source sentence and one target-language translation (the FLORES reference translation itself there is no separate machine-generated hypothesis being checked against a reference; the translation *is* what’s being judged).

**Judges.** | | GLM-4 (9B) | Qwen2.5 (7B) | Serving | Local, via Ollama | Local, via Ollama | | Cost | Free | Free | | Role in this study | Judge 1 (primary) | Judge 2 (comparison / multi-judge partner) |

Both judges received an identical prompt (MQM-style rubric) asking for (a) a flat 0–100 score and (b) a list of specific errors, each tagged with a category and a severity (minor / major / critical). Temperature was fixed at 0 for reproducibility.

**Severity-weighted scoring.** Computed deterministically in Python from each judge’s error list — the LLM is not trusted to do the arithmetic. Weights follow the MQM convention used in WMT human evaluation (Freitag et al., 2021): minor = 1, major = 5, critical = 25 points deducted from a 100-point ceiling, floored at 0.

**Automatic metric.** LaBSE cross-lingual embedding cosine similarity between the English source and the translation, scaled to 0–100 — a reference-free adequacy proxy, in the same family as COMET-QE/CometKiwi, chosen because it requires no gated model or paid API and runs on a MacBook Air.

**Sample sizes.** All four judge × language combinations were run on the full 200-row set. Two data-quality caveats affect the numbers below and are addressed in Limitations: (1) the automatic metric only completed successfully for a minority of rows per file (8–40%); (2) one early Mandarin/GLM run file was missing the judge\_model column, but the scores themselves were complete and are included.

## Mandarin Results (English → Mandarin, n=200 per judge)

| Metric                                              | GLM-4       | Qwen2.5     |
|-----------------------------------------------------|-------------|-------------|
| Flat score — mean (std)                             | 97.6 (1.2)  | 96.2 (2.0)  |
| Flat score — min                                    | 95.0        | 85.0        |
| Severity-weighted — mean (std)                      | 99.6 (1.4)  | 99.4 (1.5)  |
| Severity-weighted — min                             | 90.0        | 94.0        |
| % sentences scored 100 (severity-weighted)          | 82.5%       | 74.0%       |
| Errors found — minor / major / critical             | 28 / 12 / 0 | 35 / 18 / 0 |
| Mean(flat – severity-weighted)                      | –1.9        | –3.2        |
| Correlation: flat vs. severity-weighted (Pearson r) | 0.52        | 0.53        |

**Reading this:** both judges rate Mandarin translations very highly and with low variance under flat scoring 98–100% of sentences score  $\geq 95$  flat, which leaves almost no room to distinguish a genuinely flawless sentence from one with a couple of real issues. Severity weighting recovers some of that lost signal: it still gives most sentences a perfect 100, but pulls flawed sentences down to 90–94, a gap flat scoring never produces. Neither judge tagged a single **critical** error anywhere in the Mandarin set consistent with Mandarin being a high-resource language both models handle comfortably; the errors found were real but recoverable (mistranslated qualifiers, minor omissions, word-order shifts), not meaning-reversing mistakes.

**Inter-judge agreement (Mandarin):** Pearson  $r = 0.35$  (severity-weighted scores),  $r = 0.06$  (flat scores) i.e. the two judges agree *more* once severity weighting strips out flat-score noise, but agreement is still moderate at best.

## Cantonese Results (English → Cantonese, n=200 per judge)

| Metric                                              | GLM-4       | Qwen2.5     |
|-----------------------------------------------------|-------------|-------------|
| Flat score — mean (std)                             | 95.8 (1.6)  | 95.1 (3.0)  |
| Flat score — min                                    | 85.0        | 65.0        |
| Severity-weighted — mean (std)                      | 98.9 (2.0)  | 98.9 (2.9)  |
| Severity-weighted — min                             | 90.0        | 75.0        |
| % sentences scored 100 (severity-weighted)          | 53.0%       | 65.5%       |
| Errors found — minor / major / critical             | 90 / 26 / 0 | 47 / 24 / 2 |
| Mean(flat – severity-weighted)                      | –3.1        | –3.8        |
| Correlation: flat vs. severity-weighted (Pearson r) | 0.35        | 0.74        |

**Reading this:** Cantonese scores lower and more variably than Mandarin under both judges and both scoring methods expected, given Cantonese is comparatively under-represented in both models’ training data. GLM found **more total errors** (90 minor + 26 major) than on Mandarin, but never escalated any of them to critical. Qwen found fewer total errors but is the only judge that used the critical label at all both critical flags were in Cantonese (see Error Coverage section below for what they were). Qwen’s flat-score minimum of 65 (vs. GLM’s 85) shows Qwen is willing to

express low confidence in its holistic score even when it doesn't always back that up with a matching severity-weighted deduction worth watching if Qwen is used standalone.

**Inter-judge agreement (Cantonese):** Pearson  $r = 0.23$  (severity-weighted),  $r = 0.36$  (flat) lower than Mandarin's severity-weighted agreement (0.35), confirming the two judges see Cantonese translation quality somewhat differently from each other, which is exactly the situation multi-judge setups are meant to help with.

## **Error Coverage and Qualitative Analysis (Multi-Judge vs. Single Judge)**

For every sentence, a judge is counted as having “flagged” it if it tagged at least one **major or critical** error (minor-only sentences are excluded from this count, since minor issues rarely change a go/no-go quality decision).

|                                                          | Mandarin (n=200)          | Cantonese (n=200)          |
|----------------------------------------------------------|---------------------------|----------------------------|
| Flagged by GLM only                                      | 6                         | 14                         |
| Flagged by Qwen only                                     | 13                        | 19                         |
| Flagged by both                                          | 5                         | 7                          |
| Flagged by neither                                       | 176                       | 160                        |
| <b>Any single judge alone</b> (GLM or Qwen individually) | 5.5% – 9.0% of sentences  | 10.5% – 13.0% of sentences |
| <b>Multi-judge union</b> (either judge)                  | <b>12.0%</b> of sentences | <b>20.0%</b> of sentences  |

In both languages, combining the two judges roughly **doubles** the share of sentences flagged with a real (major/critical) issue compared to relying on either judge alone. The effect is larger in absolute terms for Cantonese (+7 to +9.5 percentage points) than Mandarin (+3 to +6.5 points), directly supporting the acceptance criterion asking whether multi-judge setups catch more Cantonese-specific errors.

## **Concrete examples errors one judge caught and the other missed (Cantonese)**

**Caught by Qwen, missed by GLM:** - *Row 31* — EN: “In 1995 he was voted the best player in **Partizan’s** history.” Translation: “1995年，佢獲選為**游擊隊**歷史上最佳球員。” Qwen correctly flagged that “Partizan” (the football club’s proper name) was mistranslated as “游擊隊” (literally “guerrilla force”/partisan fighters) — a genuine proper-noun error present in the FLORES reference translation itself, not something a native Cantonese speaker would catch by skimming. GLM did not flag this sentence at all. - *Row 77* — flagged a non-standard astronomy term (“光球層” vs. the standard “光球” for “photosphere”) that GLM missed.

**Caught by GLM, missed by Qwen:** - *Row 32* — flagged “太陽馬戲團” as an imprecise rendering of “Cirque du Soleil” — a more debatable catch, since that Chinese name is in fairly common use, illustrating that not every judge-flagged “error” is unambiguous and some would benefit from human adjudication (see Limitations). - *Row 12* — GLM flagged a real accuracy issue: “was one of the four killed” was translated in a way that overstates certainty about which group of victims he belonged to.

**Take-away:** the two judges’ misses are not random noise on the same errors — they are catching genuinely different things, including at least one clear, unambiguous proper-noun mistranslation

(Row 31) that only showed up because a second judge was run at all. This is the strongest evidence in this study for running Cantonese through more than one judge.

### **Automatic Metric: Agreement and Disagreement**

The LaBSE reference-free similarity metric was run alongside both judges as an independent signal. **Coverage was incomplete** (see Limitations) the figures below are based only on the subset of rows where it returned a value. (*n.s.* = *not statistically significant at  $p < 0.05$  given the small available sample.*)

|                                        | Mandarin /<br>GLM | Mandarin /<br>Qwen | Cantonese /<br>GLM | Cantonese /<br>Qwen |
|----------------------------------------|-------------------|--------------------|--------------------|---------------------|
| Rows with a metric value               | 79 / 200 (40%)    | 28 / 200 (14%)     | 35 / 200 (18%)     | 16 / 200 (8%)       |
| Metric mean (std)                      | 92.2 (14.2)       | 96.8 (13.0)        | 80.6 (21.1)        | 97.1 (8.7)          |
| Spearman(flat_score, metric)           | -0.07 (n.s.)      | 0.26 (n.s.)        | 0.04 (n.s.)        | -0.14 (n.s.)        |
| Spearman(severity-weighted,<br>metric) | -0.13 (n.s.)      | 0.05 (n.s.)        | 0.23 (n.s.)        | -0.07 (n.s.)        |

**None of these correlations reach statistical significance.** At face value that suggests the judge scores and the automatic metric are measuring related but distinct things — which is expected, since the judge is scoring translation *quality* (accuracy, completeness, fluency) while LaBSE similarity is scoring cross-lingual *semantic closeness*, a related-but-different notion. Given the small effective sample sizes (as few as 16 rows), these correlation estimates carry wide uncertainty and should not be treated as a firm verdict on judge-metric relationship either way.

**Where the two clearly disagreed**, the pattern was consistent: several sentences that both judges scored as flawless (severity-weighted = 100) received low LaBSE similarity scores (e.g. Mandarin row 10: judge=100, metric=34; Cantonese row 7: judge=100, metric=-2). Spot-checking these by hand, the translations read as accurate — this looks more like a **metric weakness on short sentences with reordering** (a known limitation of sentence-embedding similarity) than a judge blind spot, but this is a manual read on a handful of cases, not a systematic check.

### **Final Recommendation**

**Scoring method — use severity-weighted scoring as the primary score, for both languages.** Flat scoring compresses almost the entire dataset into a 2–3 point band and barely separates clean sentences from flawed ones. Severity weighting keeps the same ceiling for genuinely clean translations but produces a meaningful, auditable drop for sentences with real issues, with every point loss traceable to a specific tagged error. Keep the flat score only as a secondary sanity-check number, not the number decisions get made on.

**Judge GLM alone for Mandarin; GLM + Qwen together (multi-judge) for Cantonese.** For Mandarin, GLM is cheaper to run, has similar error yield to Qwen, and the two judges agree with each other reasonably well ( $r=0.35$  on severity-weighted scores); single-judge GLM is an acceptable default given typical quality-monitoring budgets. For Cantonese, the two judges agree with each other *less* ( $r=0.23$ ), a single judge — either one — misses roughly half of the major/critical-flagged sentences the other judge would have caught, and the only two critical-severity errors in the entire study were Cantonese sentences that only Qwen caught. Cantonese is

exactly the case the acceptance criteria anticipated, and the data backs that concern up: run both judges for Cantonese and treat the union of their flags as the review queue.

**Automatic metric keep it but fix reliability before trusting its verdicts.** LaBSE similarity is free, fast, and reference-free, and the disagreement cases it did surface were worth reading. But at 8–40% coverage it isn’t yet a dependable second opinion. Until fixed, use it only as a lightweight triage flag (route judge-says-perfect / metric-says-low-similarity rows to human spot-check) rather than as evidence on its own.

**Next step before wider rollout: add a small human-labeled gold set** (30–50 sentences per language, hand-scored by a bilingual reviewer) to calibrate both judges’ severity tagging and resolve ambiguous calls like the “Cirque du Soleil” case above. This is the one acceptance-criteria item this round could not address at all.

## **Limitation and Interpretation**

**Automatic metric coverage was incomplete and the cause wasn’t diagnosed.** Between 60% and 92% of rows per file returned no LaBSE similarity value. The failures were scattered throughout each run rather than clustered at the start or end, which points to an intermittent runtime issue (most likely memory pressure from running an embedding model repeatedly on an 8–16GB laptop alongside the judge model) rather than a systematic content problem — but this wasn’t confirmed, since the original run swallowed the underlying exception instead of logging it. Any conclusion drawn from the automatic-metric section should be treated as directional, not definitive, until this is fixed and the metric is re-run at full coverage.

**Small critical-error sample.** Only 2 critical errors were tagged across 800 judge-sentence evaluations (400 sentences  $\times$  2 judges). That’s too few to draw statistically firm conclusions about *which* judge is better at catching critical errors — it’s suggestive, not proof, that Qwen is more willing to use that label.

**Two judges, not three.** The acceptance criteria allowed for two or three judges; this round tested two (GLM, Qwen). A third judge would let the “flagged by 2 of 3” signal be used as a higher-confidence filter than “flagged by either of 2,” which single-pair agreement can’t distinguish from one judge simply being noisier.

**Reference-free metric is a proxy, not true COMET.** LaBSE cosine similarity was used in place of the actual COMET-QE/CometKiwi model to avoid a gated-model / Hugging Face login requirement. It’s a reasonable free stand-in but is not literally the metric named in the acceptance criteria; swapping in real CometKiwi later (code already stubbed for this) would let the automatic-metric section be re-validated against a more standard, published metric.

**One data file inconsistency.** `mandarin_glm4_judge_results.csv` was missing the `judge_model` column present in the other three files (a pipeline versioning artifact from before that column was added). All score values in that file are intact and were used normally; only the column is missing, with no effect on the numbers reported here.

## **Conclusion**

| # | Criterion                                                                                         | Status                                                                                                                                                                                            |
|---|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Test GLM on English→Mandarin, compare vs. current judge                                           | <b>Done.</b> Compared against flat scoring (the pre-existing method) and against Qwen as an independent second judge.                                                                             |
| 2 | Test GLM on English→Cantonese, compare vs. current judge and human review where possible          | <b>Partially done.</b> Judge comparison complete; human-review comparison <b>not possible</b> — no human-labeled data available. Flagged as the top next step.                                    |
| 3 | Test combined multi-judge (2–3 judges) vs. single-judge setup                                     | <b>Done</b> for 2 judges (GLM + Qwen). A third judge was not tested.                                                                                                                              |
| 4 | Record whether multi-judge catches errors — especially Cantonese-specific — a single judge misses | <b>Done.</b> Multi-judge roughly doubles error-flagged coverage in both languages, with a larger absolute gain in Cantonese; concrete examples documented.                                        |
| 5 | Design and test severity-weighted scoring vs. flat score                                          | <b>Done.</b> MQM-style weighting implemented, computed deterministically, and compared against flat scores for both judges and both languages.                                                    |
| 6 | Add $\geq 1$ reference-free automatic metric, note agreement/disagreement                         | <b>Done, with a caveat.</b> LaBSE similarity implemented and compared; metric coverage was unreliable (8–40%) and should be fixed before the agreement/disagreement numbers are treated as final. |
| 7 | Document one clear final recommendation                                                           | <b>Done</b> see Final Recommendation section.                                                                                                                                                     |